Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years developing customer-focused applications and cloud services across the full software lifecycle. Experienced in C++, Java, Python, TypeScript, object-oriented design, Azure, SQL, automated testing, debugging, Git, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected maintainable Azure SQL and Blob Storage services for 500+ users, refining normalized schemas, indexed queries, and secure access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Refactored fault-tolerant background workflows for 1,000+ jobs using explicit states, retries, and failure isolation, tripling concurrent throughput while improving code maintainability and reliable ongoing production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask services for secure upload, status, history, and grading workflows, translating instructor requirements into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated validation, error handling, and AWS RDS persistence, reducing turnaround 60% while maintaining stable application behavior during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible data and simulation pipelines processing 100,000+ trajectory records, applying structured testing, debugging, and performance analysis to improve long-horizon planning accuracy 18% across controlled evaluations.
- Collaborated on model design reviews and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop stability 15% across repeated highly controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented API and ETL components using reusable design patterns, AWS Glue, and S3, improving response times 30% and pipeline efficiency 68% across complex cross-platform customer workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving engineering confidence in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular MERN services and WebSocket synchronization across eight resource, customer, billing, and scheduling workflows, supporting 2 million+ Azure SQL records with clear independently maintainable service boundaries.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving overall software quality and consistent operational reliability.

Skills

Languages & Fundamentals: C++, Java, Python, JavaScript, TypeScript, SQL, Bash, object-oriented programming, data structures, algorithms
Applications & APIs: React, Node.js, Flask, FastAPI, REST APIs, WebSockets, microservices, responsive UI, requirements analysis, software lifecycle
Data & Cloud: Azure, AWS, Azure SQL, SQL Server, PostgreSQL, MySQL, MongoDB, S3, Blob Storage, data modeling, indexing, query optimization
Quality & Delivery: Git, GitHub Actions, Jenkins, CI/CD, Docker, PyTest, Mockito, unit testing, integration testing, debugging, code review, Agile, Scrum, production support

Projects

CPU-Scheduling Simulator | *C++, Simulation, Operating Systems, React* | Live
- Built a C++ simulator modeling FCFS, SJF, SRT, Round Robin, priority, and multilevel queue scheduling with repeatable workloads, reducing average turnaround 25% across 1,000+ process configurations.

Chat2SQL | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker* | Code
- Built a cross-platform workspace supporting 10+ relational and NoSQL storage systems, with schema-aware query generation, structured validation, optimized execution, and Docker-backed deployment across Java and TypeScript services.

FinNexus | *TypeScript, WebSockets, MongoDB, Kubernetes, CI/CD* | Live
- Built a scalable application supporting 50,000+ users and one million transactions, integrating secure APIs, real-time synchronization, advanced filtering, MongoDB persistence, and containerized CI/CD deployment workflows.